Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 32

The Principle of Inclusion-Exclusion

Covers Chartrand & Zhang, Section 8.2

Status of this set

Ungraded practice. Solutions released with it. Show the full expression, not just the final number.

Part I: How to Practice This Material

The two formulas

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Signs alternate so that every Venn region is counted exactly once.

Two habits

Count the complement when the direct count is messy: $|\overline{A}| = |U| - |A|$.
For divisibility, $|A_a \cap A_b|$ counts multiples of the least common multiple — which is ab only when a and b share no factor.

Three mistakes to expect

1. **Omitting $+|A \cap B \cap C|$.** The answer comes out too small.
2. **Using ab for the pairwise intersection** when a and b share a factor.
3. **Adding when the sets overlap.** That is what this principle exists to correct.

Part II: Practice Problems

Section A — Two sets

1. How many integers from 1 to 1000 are divisible by 4 or by 6?

2. How many bit strings of length 8 begin with 11 or end with 00?

Section B — Three sets

3. How many integers from 1 to 200 are divisible by 2, 3, or 7?

4. A survey of 100 moviegoers found 60 buy popcorn, 45 buy candy, 55 buy a soft drink, 25 buy popcorn and candy, 30 buy popcorn and a drink, 20 buy candy and a drink, and 10 buy all three.

(a) How many buy at least one of the three?

(b) How many buy none of them?

Section C — Complement counting

5. How many integers from 1 to 300 are divisible by none of 2, 3, 5?

Section D — Writing

6. In four to six complete sentences, explain why the three-set formula alternates signs. Track a single element that lies in all three sets, and say how many times each term counts it.